Biz and Hypotheses

Business Proposition: Personalized Music Insights and Optimization Tool

This tool aims to leverage streaming data insights to help artists, producers, and streaming platforms understand the factors influencing music popularity. The objectives are threefold:

1. Optimize Song Structures: By analyzing how the average duration of a song influences its popularity, the tool will provide recommendations on optimal song lengths to maximize audience engagement and streaming numbers.

2. Analyze the Impact of Explicit Content: By examining the influence of explicit lyrics on song popularity over time, the tool will help stakeholders understand consumer preferences and adapt their content strategies accordingly.

3. Explore Regional Genre Preferences (additional insight): Understand if certain music genres are more popular in specific regions, enabling platforms to tailor content offerings to maximize engagement.

The insights derived from these analyses will guide stakeholders in making data-driven decisions, enhancing content delivery that resonates with diverse audiences.

Potential Business Names:

LyricaLytics: A catchy blend of "Lyrics" and "Analytics," particularly fitting if we're focusing on analyzing explicit content and song structure.

Melody Metrics: This title captures both the musical aspect ("Melody") and the analytical focus ("Metrics") of our business.

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Dataset: Spotify tracks chart dataset (2014-2022)
https://www.kaggle.com/datasets/jfreyberg/spotify-chart-data/data

- This dataset is comprised of weekly Spotify track chart data from 2014 to 2022.

- The dataset is obtained from https://kworb.net/spotify/ and the official Spotify API. In contrast to other open datasets, this one includes information on the artists, i.e. names and genre, with precise correlations also used by Spotify (such as standardized genre names).

- Additionally, the dataset includes the same track multiple times if it appeared in the Spotify charts multiple times, each time with the respective metadata, e.g. chart position and number of streams.

Hypothesis 1: Impact of Song Duration on Popularity

The average stream duration of a song influences its total number of streams, indicating which song lengths were the most appealing to listeners between the years of 2014-2022.

- Goal: Examine the relationship between the average stream duration and total streams to determine if there's an optimal song length that maximizes listenership. Find the perfect song duration for the business investment!

- Why it's interesting: Understanding which song durations tend to attract more streams can help artists and producers decide on song structures that are more likely to succeed on streaming platforms.

To test this hypothesis:

1. Data Preparation:
   - Use `Duration (Min)` vs `Streams` to identify patterns.

2. Analysis:
   - Use SQL or a visualization tool to create a scatter plot to visualize the relationship between the average stream duration and the total or monthly streams.
   - Look for trends such as whether shorter, medium, or longer songs are generally more popular.

3. Insights:
   - Identify if there's a specific range of durations that tends to correlate with higher stream counts.

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Hypothesis 2: Impact of Explicit Lyrics on Song Popularity

Songs with explicit lyrics have a significant impact on streaming popularity throughout the years, with trends showing changes in consumer preferences.

- Goal: Investigate the correlation between explicit lyrics and song popularity, analyzing if these songs rank up more streaming power. In other words, has content become more explicit over the years, thereby generating a larger audience? Does explicit content really sell better?

- Why it's interesting: Understanding the impact of explicit content helps platforms and artists predict consumer trends and tailor their content strategies for audience appeal.

To test this hypothesis:

1. Data Preparation: Compare the `Explicit` boolean indicator in our dataset with `Streams`.

2. Analysis:
   - Use SQL to segment songs by explicit content and analyze their stream counts.
   - Check for trends over time using a temporal analysis.

3. Insights:
   - Determine if there is a trend in the popularity of songs with explicit content and explore what this could mean for content strategy.

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Additional Insight:

Analyze regional genre preferences to determine if certain music genres are more popular in specific regions, using columns like `Country` or `Region`.

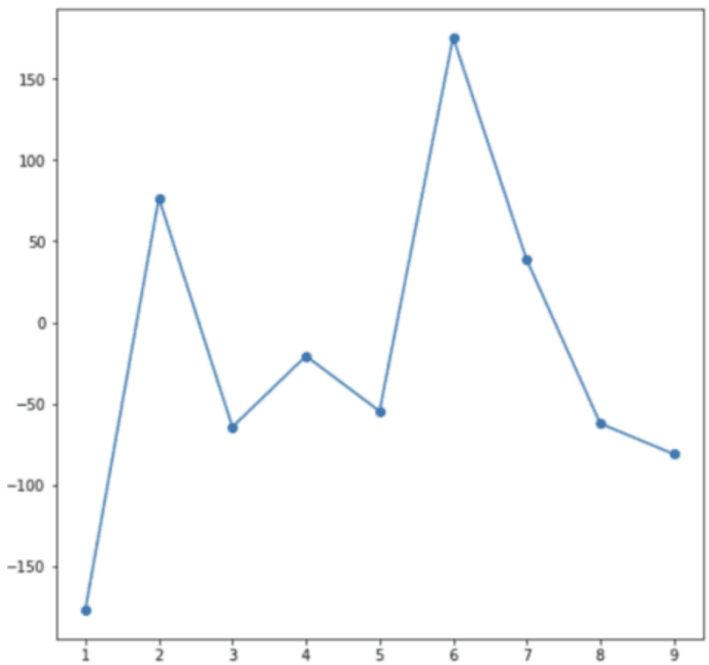
Data Visualization Approaches

The best columns to use in the validation of our hypotheses:

Hypothesis 1: Impact of Song Duration on Popularity

Columns to Use:
1. ‘duration’: Represents the song's length in milliseconds. Convert this to minutes for easier interpretation in visualizations!
2. ‘streams’: Indicates the total number of times a song has been streamed. This measures the song's popularity.

- Scatter plot to visualize the relationship, where one axis (X) represents ‘duration’ in minutes, and the other (Y) represents ‘streams’.




```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load your dataset
df = pd.read_csv('path_to_your_file/spotify_chart_data.csv')

# Convert `duration` from milliseconds to minutes
df['duration_min'] = df['duration'] / 60000
```

```
# Plotting the scatter plot
plt.figure(figsize=(10, 6))
sns.scatterplot(x='duration_min', y='streams', data=df, alpha=0.6)
plt.title('Song Duration vs Total Streams')
plt.xlabel('Average Stream Duration (Minutes)')
plt.ylabel('Total Streams')
plt.grid(True)
plt.show()
```

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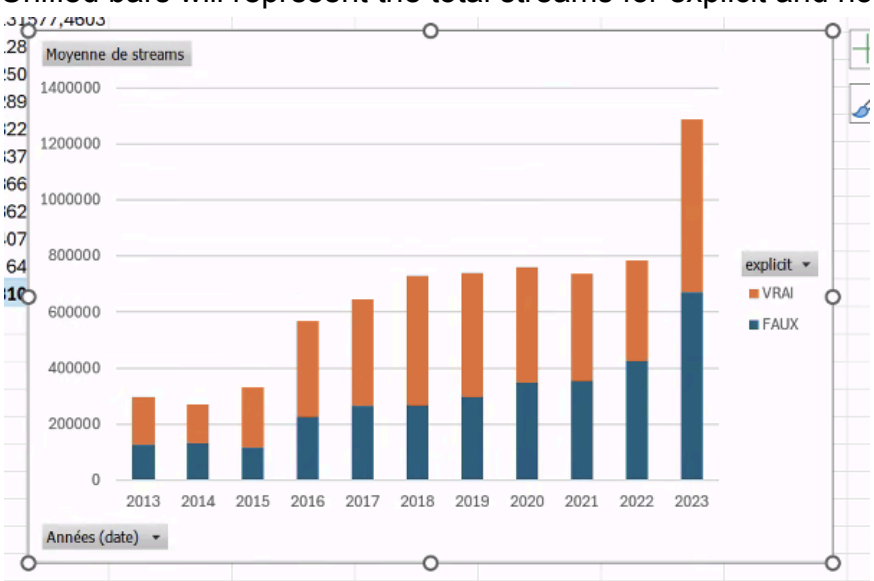
Hypothesis 2: Impact of Explicit Lyrics on Song Popularity

Columns to Use:
1. ‘explicit’: A boolean indicating whether a song has explicit lyrics.
2. ‘streams’: Again, this measures the popularity of the song, allowing us to compare total streams between explicit and non-explicit songs.
3. ‘date’: Allows for the temporal analysis of explicit content trends over time.

<u>Bar chart,</u> where one axis stands for ‘number of streams’ and the other stands for ‘year’. Unified bars will represent the total streams for explicit and non-explicit songs.



- Use the ‘explicit’ column to categorize songs into two groups: explicit and non-explicit.
- Calculate the proportion of explicit songs per year by grouping the data by year and taking the mean of the ‘explicit’ column. This

provides the proportion of songs that are labeled as explicit for each year.

- Calculate the total streams for each category. This allows us to see which type (explicit or non-explicit) has been streamed more.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Assume 'df' is your DataFrame loaded from the dataset
df = pd.read_csv('path_to_your_file/spotify_chart_data.csv')

# Calculate the total streams for explicit and non-explicit songs per
year
stream_summary = df.groupby(['year',
'explicit'])['streams'].sum().unstack().fillna(0)

# Rename columns for clarity
stream_summary.columns = ['Non-Explicit', 'Explicit']

# Calculate the proportion of explicit songs per year
explicit_proportion = df.groupby('year')['explicit'].mean()

# Plotting the bar chart for streams
plt.figure(figsize=(12, 6))
stream_summary.plot(kind='bar', stacked=True, ax=plt.gca(),
color=['skyblue', 'salmon'])
plt.title('Total Streams of Explicit vs Non-Explicit Songs by Year')
plt.xlabel('Year')
plt.ylabel('Total Streams')
plt.legend(title='Song Type')
plt.tight_layout()
plt.grid(axis='y')

# Overlay a line for the proportion of explicit songs
plt.twinx()
plt.plot(explicit_proportion.index, explicit_proportion.values,
color='green', marker='o', linestyle='-')
plt.ylabel('Proportion of Explicit Songs')
```

```
plt.show()
```

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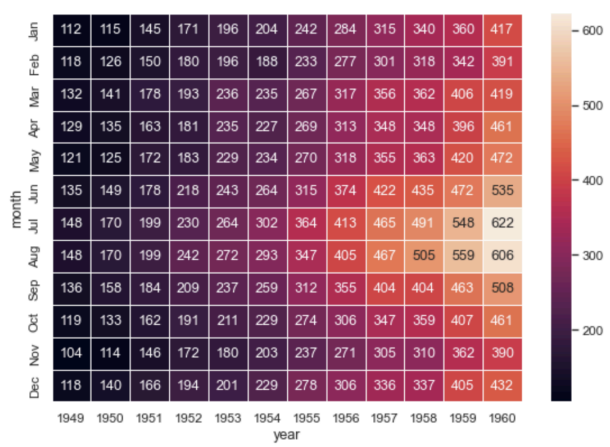
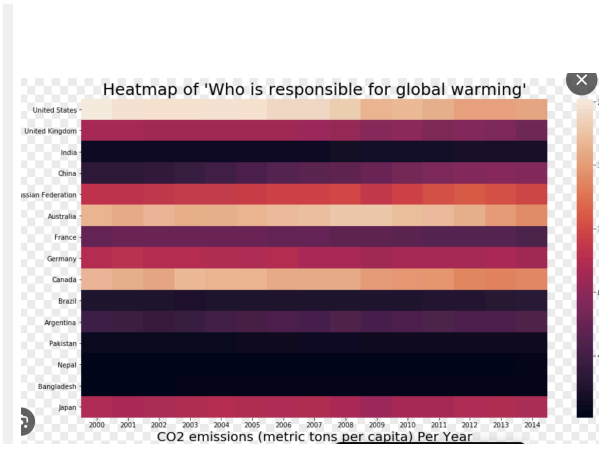
Additional Insight: Regional Genre Preferences

Columns to Use:
1. ‘country’: Indicates the country/region for regional analysis.
2. ‘artist_genres’: Provides the genre(s) associated with the artists. We can choose to extract the primary genre for simplification.
3. ‘streams’: Allows us to compare popular music genres based on stream counts in different regions.

Heat map to display genre-streaming intensities across various countries.





- Ensure that artist_genres is parsed correctly, consider extracting the primary genre.
- Each cell in the map shows the intensity of streams for a particular genre in a specific country.
- Darker shades denote higher streaming numbers, indicating greater popularity.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load your dataset
df = pd.read_csv('path_to_your_file/spotify_chart_data.csv')

# Assuming genres are semicolon-separated, extract the primary genre
```

```
df['primary_genre'] = df['artist_genres'].apply(lambda x: x.split(';')[0] if
pd.notna(x) and ';' in x else x)

# Create a pivot table for streams by country and genre
region_genre_data = df.pivot_table(values='streams', index='country',
columns='primary_genre', aggfunc='sum').fillna(0)

# Plotting the heat map
plt.figure(figsize=(14, 8))
sns.heatmap(region_genre_data, cmap='Blues', cbar_kws={'label': 'Total
Streams'}, annot=False)
plt.title('Genre Popularity by Country')
plt.xlabel('Genre')
plt.ylabel('Country')
plt.show()
```